

## Chapter 1

# Introduction To Flying

### Introduction

The Pilot's Handbook of Aeronautical Knowledge provides basic knowledge for the student pilot learning to fly, as well as pilots seeking advanced pilot certification. For detailed information on a variety of specialized flight topics, see specific Federal Aviation Administration (FAA) handbooks and Advisory Circulars (ACs).

This chapter offers a brief history of flight, introduces the history and role of the FAA in civil aviation, FAA regulations and standards, government references and publications, eligibility for pilot certificates, available routes to flight instruction, the role of the Certificated Flight Instructor (CFI) and Designated Pilot Examiner (DPE) in flight training, Practical Test Standards (PTS), and new, industry-developed Airman Certification Standards (ACS) framework that will eventually replace the PTS.



Information Manual (AIM)  
The Information Manual is designed to provide  
community with basic flight information.  
Flying Handbook

Plane Flying Handbook is designed as a technical  
to introduce basic flight information.  
Aviation Instructor's Handbook

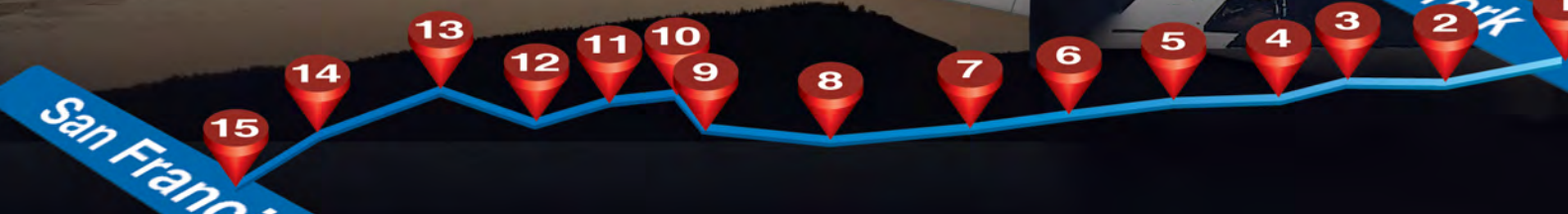
Aviation Instructor's Handbook provides the foundation  
beginning instructors to understand and apply the  
Instrument Flying Handbook

The Instrument Flying Handbook is designed for use by  
instrument flight instructors and pilots preparing for  
instrument flight

Instrument Procedures Handbook  
The Instrument Procedures Handbook is designed as a  
technical reference for professional pilots who operate  
under IFR in the NAS and expands on information contained  
in the Instrument Flying Handbook.



| Code of Federal Regulations |                                                                                                                                                                                                          |
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| Aeronautics and Space       |                                                                                                                                                                                                          |
| Subchapters                 | Chapter 1. Federal Aviation Administration                                                                                                                                                               |
| A                           | Definitions (definitions and abbreviations)                                                                                                                                                              |
| B                           | Procedural rules (rulemaking process, claims, enforcement)                                                                                                                                               |
| C                           | Aircraft (Aircraft certification procedures, Airworthiness standards [parts 25 through 39], depending on type of aircraft], airworthiness directives [39], maintenance [43], aircraft registration [47]) |
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## History of Flight

From prehistoric times, humans have watched the flight of birds, and longed to imitate them, but lacked the power to do so. Logic dictated that if the small muscles of birds can lift them into the air and sustain them, then the larger muscles of humans should be able to duplicate the feat. No one knew about the intricate mesh of muscles, sinew, heart, breathing system, and devices not unlike wing flaps, variable-camber and spoilers of the modern airplane that enabled a bird to fly. Still, thousands of years and countless lives were lost in attempts to fly like birds.

The identity of the first “bird-men” who fitted themselves with wings and leapt off of cliffs in an effort to fly are lost in time, but each failure gave those who wished to fly questions that needed to be answered. Where had the wing flappers gone wrong? Philosophers, scientists, and inventors offered solutions, but no one could add wings to the human body and soar like a bird. During the 1500s, Leonardo da Vinci filled pages of his notebooks with sketches of proposed flying machines, but most of his ideas were flawed because he clung to the idea of birdlike wings. [Figure 1-1] By 1655, mathematician, physicist, and inventor Robert Hooke concluded that the human body does not possess the strength to power artificial wings. He believed human flight would require some form of artificial propulsion.

The quest for human flight led some practitioners in another direction. In 1783, the first manned hot air balloon, crafted by Joseph and Etienne Montgolfier, flew for 23 minutes. Ten days later, Professor Jacques Charles flew the first gas balloon. A madness for balloon flight captivated the public’s imagination and for a time flying enthusiasts turned their expertise to the promise of lighter-than-air flight. But for all its majesty in the air, the balloon was little more than a

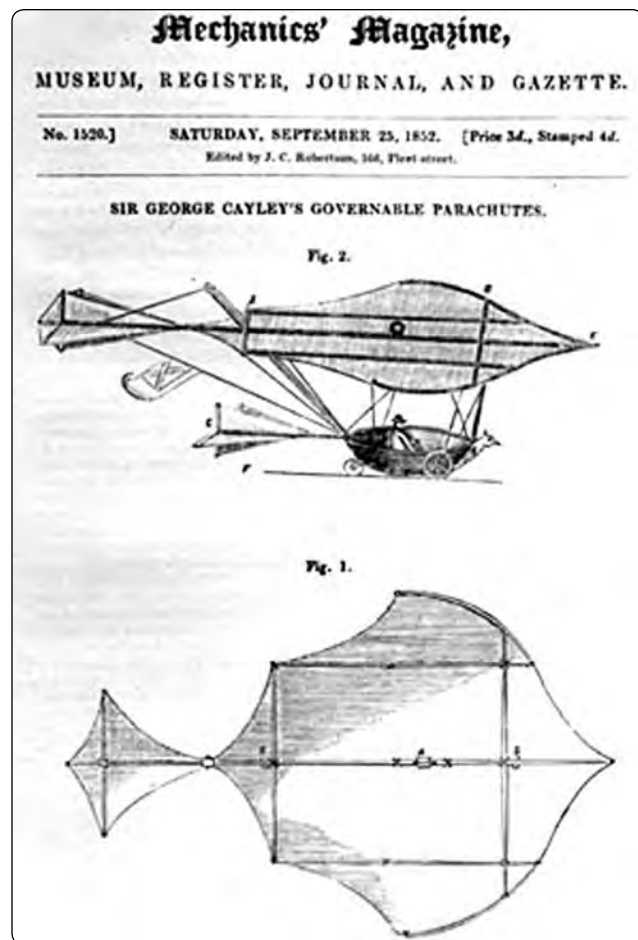


**Figure 1-1.** Leonardo da Vinci’s ornithopter wings.

billowing heap of cloth capable of no more than a one-way, downwind journey.

Balloons solved the problem of lift, but that was only one of the problems of human flight. The ability to control speed and direction eluded balloonists. The solution to that problem lay in a child’s toy familiar to the East for 2,000 years, but not introduced to the West until the 13th century—the kite. The kites used by the Chinese for aerial observation, to test winds for sailing, as a signaling device, and as a toy, held many of the answers to lifting a heavier-than-air device into the air.

One of the men who believed the study of kites unlocked the secrets of winged flight was Sir George Cayley. Born in England 10 years before the Montgolfier balloon flight, Cayley spent his 84 years seeking to develop a heavier-than-air vehicle supported by kite-shaped wings. [Figure 1-2] The “Father of Aerial Navigation,” Cayley discovered the basic principles on which the modern science of aeronautics is founded; built what is recognized as the first successful flying model; and tested the first full-size man-carrying airplane.



**Figure 1-2.** Glider from 1852 by Sir George Cayley, British aviator (1773–1857).



For the half-century after Cayley's death, countless scientists, flying enthusiasts, and inventors worked toward building a powered flying machine. Men, such as William Samuel Henson, who designed a huge monoplane that was propelled by a steam engine housed inside the fuselage, and Otto Lilienthal, who proved human flight in aircraft heavier than air was practical, worked toward the dream of powered flight. A dream turned into reality by Wilbur and Orville Wright at Kitty Hawk, North Carolina, on December 17, 1903.

The bicycle-building Wright brothers of Dayton, Ohio, had experimented for 4 years with kites, their own homemade wind tunnel, and different engines to power their biplane. One of their great achievements in flight was proving the value of the scientific, rather than a build-it-and-see approach. Their biplane, The Flyer, combined inspired design and engineering with superior craftsmanship. [Figure 1-3] By the afternoon of December 17th, the Wright brothers had flown a total of 98 seconds on four flights. The age of flight had arrived.

### History of the Federal Aviation Administration (FAA)

During the early years of manned flight, aviation was a free for all because no government body was in place to establish policies or regulate and enforce safety standards. Individuals were free to conduct flights and operate aircraft with no government oversight. Most of the early flights were conducted for sport. Aviation was expensive and became the playground of the wealthy. Since these early airplanes were small, many people doubted their commercial value. One group of individuals believed otherwise and they became the genesis for modern airline travel.

P. E. Fansler, a Florida businessman living in St. Petersburg, approached Tom Benoist of the Benoist Aircraft Company in St. Louis, Missouri, about starting a flight route from St.

Petersburg across the waterway to Tampa. Benoist suggested using his "Safety First" airboat and the two men signed an agreement for what would become the first scheduled airline in the United States. The first aircraft was delivered to St. Petersburg and made the first test flight on December 31, 1913. [Figure 1-4]

A public auction decided who would win the honor of becoming the first paying airline customer. The former mayor of St. Petersburg, A. C. Pheil, made the winning bid of \$400.00, which secured his place in history as the first paying airline passenger.

On January 1, 1914, the first scheduled airline flight was conducted. The flight length was 21 miles and lasted 23 minutes due to a headwind. The return trip took 20 minutes. The line, which was subsidized by Florida businessmen, continued for 4 months and offered regular passage for \$5.00 per person or \$5.00 per 100 pounds of cargo. Shortly after the opening of the line, Benoist added a new airboat that afforded more protection from spray during takeoff and landing. The routes were also extended to Manatee, Bradenton, and Sarasota giving further credence to the idea of a profitable commercial airline.

The St. Petersburg-Tampa Airboat Line continued throughout the winter months with flights finally being suspended when the winter tourist industry began to dry up. The airline operated for only 4 months, but 1,205 passengers were carried without injury. This experiment proved commercial passenger airline travel was viable.

The advent of World War I offered the airplane a chance to demonstrate its varied capabilities. It began the war as a reconnaissance platform, but by 1918, airplanes were being



Figure 1-3. First flight by the Wright brothers.



Figure 1-4. Benoist airboat.

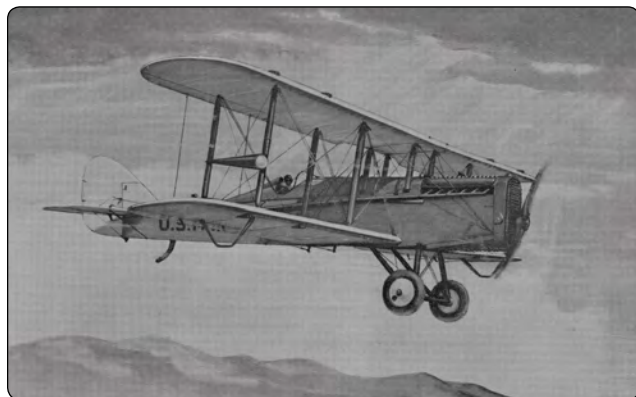
mass produced to serve as fighters, bombers, trainers, as well as reconnaissance platforms.

Aviation advocates continued to look for ways to use airplanes. Airmail service was a popular idea, but the war prevented the Postal Service from having access to airplanes. The War Department and Postal Service reached an agreement in 1918. The Army would use the mail service to train its pilots in flying cross-country. The first airmail flight was conducted on May 15, 1918, between New York and Washington, DC. The flight was not considered spectacular; the pilot became lost and landed at the wrong airfield. In August of 1918, the United States Postal Service took control of the airmail routes and brought the existing Army airmail pilots and their planes into the program as postal employees.

### Transcontinental Air Mail Route

Airmail routes continued to expand until the Transcontinental Mail Route was inaugurated. [Figure 1-5] This route spanned from San Francisco to New York for a total distance of 2,612 miles with 13 intermediate stops along the way. [Figure 1-6] On May 20, 1926, Congress passed the Air Commerce Act, which served as the cornerstone for aviation within the United States. This legislation was supported by leaders in the aviation industry who felt that the airplane could not reach its full potential without assistance from the Federal Government in improving safety.

The Air Commerce Act charged the Secretary of Commerce with fostering air commerce, issuing and enforcing air traffic rules, licensing pilots, certificating aircraft, establishing airways, and operating and maintaining aids to air navigation. The Department of Commerce created a new Aeronautics Branch whose primary mission was to provide oversight for the aviation industry. In addition, the Aeronautics Branch took over the construction and operation of the nation's system of lighted airways. The Postal Service, as part of the Transcontinental Air Mail Route system, had initiated this system. The



**Figure 1-5.** The de Havilland DH-4 on the New York to San Francisco inaugural route in 1921.



**Figure 1-6.** The transcontinental airmail route ran from New York to San Francisco.

Department of Commerce made significant advances in aviation communications, including the introduction of radio beacons as an effective means of navigation.

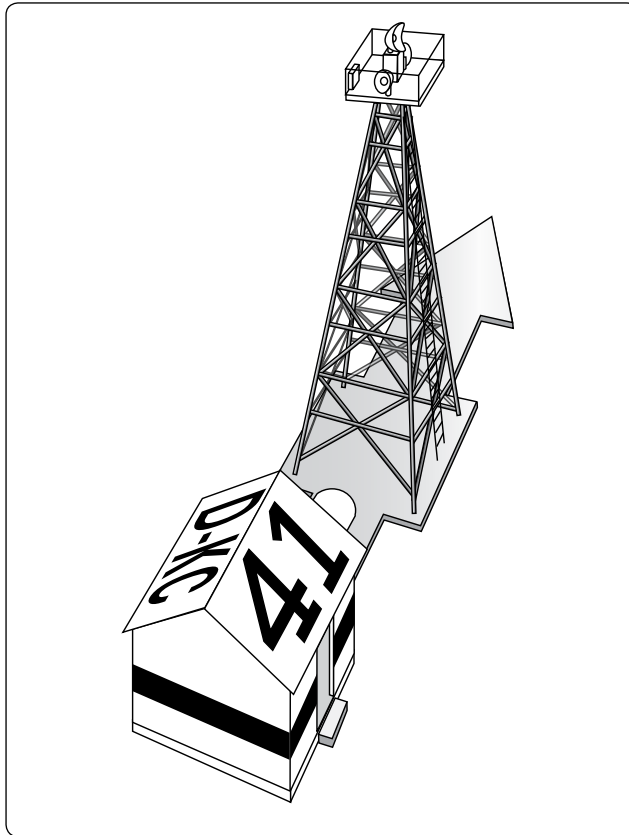
Built at intervals of approximately 10 miles apart, the standard beacon tower was 51 feet high, and was topped with a powerful rotating light. Below the rotating light, two course lights pointed forward and back along the airway. The course lights flashed a code to identify the beacon's number. The tower usually stood in the center of a concrete arrow 70 feet long. A generator shed, where required, stood at the "feather" end of the arrow. [Figure 1-7]

### Federal Certification of Pilots and Mechanics

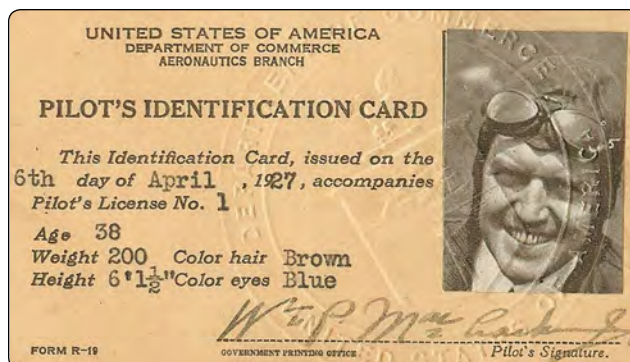
The Aeronautics Branch of the Department of Commerce began pilot certification with the first license issued on April 6, 1927. The recipient was the Chief of the Aeronautics Branch, William P. MacCracken, Jr. [Figure 1-8] (Orville Wright, who was no longer an active flier, had declined the honor.) MacCracken's license was the first issued to a pilot by a civilian agency of the Federal Government. Some 3 months later, the Aeronautics Branch issued the first Federal aircraft mechanic license.

Equally important for safety was the establishment of a system of certification for aircraft. On March 29, 1927, the Aeronautics Branch issued the first airworthiness type certificate to the Buhl Airster CA-3, a three-place open biplane.

In 1934, to recognize the tremendous strides made in aviation and to display the enhanced status within the department, the Aeronautics Branch was renamed the Bureau of Air Commerce. [Figure 1-9] Within this time frame, the Bureau of Air Commerce brought together a group of airlines



**Figure 1-7.** A standard airway beacon tower.



**Figure 1-8.** The first pilot license was issued to William P. MacCracken, Jr.



**Figure 1-9.** The third head of the Aeronautics Branch, Eugene L. Vidal, is flanked by President Franklin D. Roosevelt (left) and Secretary of Agriculture Henry A. Wallace (right). The photograph was taken in 1933. During Vidal's tenure, the Aeronautics Branch was renamed the Bureau of Air Commerce on July 1, 1934. The new name more accurately reflected the status of the organization within the Department of Commerce.

and encouraged them to form the first three Air Traffic Control (ATC) facilities along the established air routes. Then in 1936, the Bureau of Air Commerce took over the responsibilities of operating the centers and continued to advance the ATC facilities. ATC has come a long way from the early controllers using maps, chalkboards, and performing mental math calculations in order to separate aircraft along flight routes.

### The Civil Aeronautics Act of 1938

In 1938, the Civil Aeronautics Act transferred the civil aviation responsibilities to a newly created, independent body, named the Civil Aeronautics Authority (CAA). This Act empowered the CAA to regulate airfares and establish new routes for the airlines to service.

President Franklin Roosevelt split the CAA into two agencies—the Civil Aeronautics Administration (CAA) and the Civil Aeronautics Board (CAB). Both agencies were still part of the Department of Commerce but the CAB functioned independently of the Secretary of Commerce. The role of the CAA was to facilitate ATC, certification of airmen and aircraft, rule enforcement, and the development of new airways. The CAB was charged with rule making to enhance safety, accident investigation, and the economic regulation of the airlines. Then in 1946, Congress gave the CAA the responsibility of administering the Federal Aid



Airport Program. This program was designed to promote the establishment of civil airports throughout the country.

### **The Federal Aviation Act of 1958**

By mid-century, air traffic had increased and jet aircraft had been introduced into the civil aviation arena. A series of mid-air collisions underlined the need for more regulation of the aviation industry. Aircraft were not only increasing in numbers, but were now streaking across the skies at much higher speeds. The Federal Aviation Act of 1958 established a new independent body that assumed the roles of the CAA and transferred the rule making authority of the CAB to the newly created Federal Aviation Agency (FAA). In addition, the FAA was given complete control of the common civil-military system of air navigation and ATC. The man who was given the honor of being the first Administrator of the FAA was former Air Force General Elwood Richard “Pete” Quesada. He served as the administrator from 1959–1961. [Figure 1-10]

### **Department of Transportation (DOT)**

On October 15, 1966, Congress established the Department of Transportation (DOT), which was given oversight of the transportation industry within the United States. The result was a combination of both air and surface transportation. Its mission was and is to serve the United States by ensuring a fast, safe, efficient, accessible, and convenient transportation system meeting vital national interests and enhancing the quality of life of the American people, then, now, and into



**Figure 1-10.** First Administrator of the FAA was General Elwood Richard “Pete” Quesada, 1959–1961.

the future. The DOT began operation on April 1, 1967. At this same time, the Federal Aviation Agency was renamed to the Federal Aviation Administration (FAA).

The role of the CAB was assumed by the newly created National Transportation Safety Board (NTSB), which was charged with the investigation of all transportation accidents within the United States.

As aviation continued to grow, the FAA took on additional duties and responsibilities. With the highjacking epidemic of the 1960s, the FAA was responsible for increasing the security duties of aviation both on the ground and in the air. After September 11, 2001, the duties were transferred to a newly created body called the Department of Homeland Security (DHS).

With numerous aircraft flying in and out of larger cities, the FAA began to concentrate on the environmental aspect of aviation by establishing and regulating the noise standards of aircraft. Additionally, in the 1960s and 1970s, the FAA began to regulate high altitude (over 500 feet) kite and balloon flying. In 1970, more duties were assumed by the FAA in the addition of a new federal airport aid program and increased responsibility for airport safety.

### **ATC Automation**

By the mid-1970s, the FAA had achieved a semi-automated ATC system based on a marriage of radar and computer technology. By automating certain routine tasks, the system allowed controllers to concentrate more efficiently on the vital task of providing aircraft separation. Data appearing directly on the controllers’ scopes provided the identity, altitude, and groundspeed of aircraft carrying radar beacons. Despite its effectiveness, this system required enhancement to keep pace with the increased air traffic of the late 1970s. The increase was due in part to the competitive environment created by the Airline Deregulation Act of 1978. This law phased out CAB’s economic regulation of the airlines, and CAB ceased to exist at the end of 1984.

To meet the challenge of traffic growth, the FAA unveiled the National Airspace System (NAS) Plan in January 1982. The new plan called for more advanced systems for en route and terminal ATC, modernized flight service stations, and improvements in ground-to-air surveillance and communication.

### **The Professional Air Traffic Controllers Organization (PATCO) Strike**

While preparing the NAS Plan, the FAA faced a strike by key members of its workforce. An earlier period of discord between management and the Professional Air

Traffic Controllers Organization (PATCO) culminated in a 1970 “sickout” by 3,000 controllers. Although controllers subsequently gained additional wage and retirement benefits, another period of tension led to an illegal strike in August 1981. The government dismissed over 11,000 strike participants and decertified PATCO. By the spring of 1984, the FAA ended the last of the special restrictions imposed to keep the airspace system operating safely during the strike.

### **The Airline Deregulation Act of 1978**

Until 1978, the CAB regulated many areas of commercial aviation such as fares, routes, and schedules. The Airline Deregulation Act of 1978, however, removed many of these controls, thus changing the face of civil aviation in the United States. After deregulation, unfettered free competition ushered in a new era in passenger air travel.

The CAB had three main functions: to award routes to airlines, to limit the entry of air carriers into new markets, and to regulate fares for passengers. Much of the established practices of commercial passenger travel within the United States went back to the policies of Walter Folger Brown, the United States Postmaster General during the administration of President Herbert Hoover. Brown had changed the mail payments system to encourage the manufacture of passenger aircraft instead of mail-carrying aircraft. His influence was crucial in awarding contracts and helped create four major domestic airlines: United, American, Eastern, and Transcontinental and Western Air (TWA). Similarly, Brown had also helped give Pan American a monopoly on international routes.

The push to deregulate, or at least to reform the existing laws governing passenger carriers, was accelerated by President Jimmy Carter, who appointed economist and former professor Alfred Kahn, a vocal supporter of deregulation, to head the CAB. A second force to deregulate emerged from abroad. In 1977, Freddie Laker, a British entrepreneur who owned Laker Airways, created the Skytrain service, which offered extraordinarily cheap fares for transatlantic flights. Laker’s offerings coincided with a boom in low-cost domestic flights as the CAB eased some limitations on charter flights (i.e., flights offered by companies that do not actually own planes but leased them from the major airlines). The big air carriers responded by proposing their own lower fares. For example, American Airlines, the country’s second largest airline, obtained CAB approval for “SuperSaver” tickets.

All of these events proved to be favorable for large-scale deregulation. In November 1977, Congress formally deregulated air cargo. In late 1978, Congress passed the Airline Deregulation Act of 1978, legislation that had been principally authored by Senators Edward Kennedy and

Howard Cannon. *[Figure 1-11]* There was stiff opposition to the bill—from the major airlines who feared free competition, from labor unions who feared non-union employees, and from safety advocates who feared that safety would be sacrificed. Public support was, however, strong enough to pass the act. The act appeased the major airlines by offering generous subsidies and pleased workers by offering high unemployment benefits if they lost their jobs as a result. The most important effect of the act, whose laws were slowly phased in, was on the passenger market. For the first time in 40 years, airlines could enter the market or (from 1981) expand their routes as they saw fit. Airlines (from 1982) also had full freedom to set their fares. In 1984, the CAB was finally abolished since its primary duty of regulating the airline industry was no longer necessary.

## **The Role of the FAA**

### **The Code of Federal Regulations (CFR)**

The FAA is empowered by regulations to promote aviation safety and establish safety standards for civil aviation. The FAA achieves these objectives under the Code of Federal Regulations (CFR), which is the codification of the general and permanent rules published by the executive departments and agencies of the United States Government. The regulations are divided into 50 different codes, called Titles, that represent broad areas subject to Federal regulation. FAA regulations are listed under Title 14, “Aeronautics and Space,” which encompasses all aspects of civil aviation from how to earn a pilot’s certificate to maintenance of an aircraft. Title 14 CFR Chapter 1, Federal Aviation Administration, is broken down into subchapters A through N as illustrated in *Figure 1-12*.

For the pilot, certain parts of 14 CFR are more relevant than others. During flight training, it is helpful for the pilot to become familiar with the parts and subparts that relate



**Figure 1-11.** President Jimmy Carter signs the Airline Deregulation Act in late 1978.

| Code of Federal Regulations          |        |         |             |                                                                                   |
|--------------------------------------|--------|---------|-------------|-----------------------------------------------------------------------------------|
| Title                                | Volume | Chapter | Subchapters |                                                                                   |
| Title 14<br>Aeronautics<br>and Space | 1      | I       | A           | Definitions and Abbreviations                                                     |
|                                      |        |         | B           | Procedural Rules                                                                  |
|                                      |        |         | C           | Aircraft                                                                          |
|                                      | 2      |         | D           | Airmen                                                                            |
|                                      |        |         | E           | Airspace                                                                          |
|                                      |        |         | F           | Air Traffic and General Rules                                                     |
|                                      | 3      |         | G           | Air Carriers and Operators for Compensation or Hire: Certification and Operations |
|                                      |        |         | H           | Schools and Other Certified Agencies                                              |
|                                      |        |         | I           | Airports                                                                          |
|                                      |        |         | J           | Navigational Facilities                                                           |
|                                      |        |         | K           | Administrative Regulations                                                        |
|                                      |        |         | L–M         | Reserved                                                                          |
|                                      |        |         | N           | War Risk Insurance                                                                |
|                                      | 4      | II      | A           | Economic Regulations                                                              |
|                                      |        |         | B           | Procedural Regulations                                                            |
|                                      |        |         | C           | Reserved                                                                          |
|                                      |        |         | D           | Special Regulations                                                               |
|                                      |        |         | E           | Organization                                                                      |
|                                      |        |         | F           | Policy Statements                                                                 |
|                                      |        | III     | A           | General                                                                           |
|                                      |        |         | B           | Procedure                                                                         |
|                                      |        |         | C           | Licensing                                                                         |
|                                      |        |         |             |                                                                                   |
|                                      | 5      | V       |             |                                                                                   |
|                                      |        | VI      |             |                                                                                   |
|                                      |        |         | A           | Office of Management and Budget                                                   |
|                                      |        |         | B           | Air Transportation Stabilization Board                                            |

**Figure 1-12.** Overview of 14 CFR, available online free from the FAA and for purchase through commercial sources.

to flight training and pilot certification. For instance, 14 CFR part 61 pertains to the certification of pilots, flight instructors, and ground instructors. It also defines the eligibility, aeronautical knowledge, and flight proficiency, as well as training and testing requirements for each type of pilot certificate issued. 14 CFR part 91 provides guidance in the areas of general flight rules, visual flight rules (VFR), and instrument flight rules (IFR), while 14 CFR part 43 covers aircraft maintenance, preventive maintenance, rebuilding, and alterations.

### Primary Locations of the FAA

The FAA headquarters are in Washington, DC, and there are nine regional offices strategically located across the United States. The agency's two largest field facilities are the Mike Monroney Aeronautical Center (MMAC) in Oklahoma City, Oklahoma, and the William J. Hughes Technical Center (WJHTC) in Atlantic City, New Jersey. Home to FAA training and logistics services, the MMAC provides a number of aviation safety-related and business support services. The WJHTC is the premier aviation research and development and test and evaluation facility in the country. The center's programs include testing and evaluation in ATC, communication, navigation, airports, aircraft safety, and security. Furthermore, the WJHTC is active in long-range

development of innovative aviation systems and concepts, development of new ATC equipment and software, and modification of existing systems and procedures.

### Field Offices

#### Flight Standards Service

Within the FAA, the Flight Standards Service promotes safe air transportation by setting the standards for certification and oversight of airmen, air operators, air agencies, and designees. It also promotes safety of flight of civil aircraft and air commerce by:

- Accomplishing certification, inspection, surveillance, investigation, and enforcement.
- Setting regulations and standards.
- Managing the system for registration of civil aircraft and all airmen records.

The focus of interaction between Flight Standards Service and the aviation community/general public is the Flight Standards District Office (FSDO).



### *Flight Standards District Office (FSDO)*

The FAA has approximately 80 FSDOs. [Figure 1-13] These offices provide information and services for the aviation community. FSDO phone numbers are listed in the telephone directory under Government Offices, DOT, FAA. Another convenient method of finding a local office is to use the FSDO locator available at: [www.faa.gov/about/office\\_org/field\\_offices/fsdo](http://www.faa.gov/about/office_org/field_offices/fsdo).

In addition to accident investigation and the enforcement of aviation regulations, the FSDO is also responsible for the certification and surveillance of air carriers, air operators, flight schools/training centers, and airmen including pilots and flight instructors. Each FSDO is staffed by Aviation Safety Inspectors (ASIs) who play a key role in making the nation's aviation system safe.

### **Aviation Safety Inspector (ASI)**

The ASIs administer and enforce safety regulations and standards for the production, operation, maintenance, and/or modification of aircraft used in civil aviation. They also specialize in conducting inspections of various aspects of the aviation system, such as aircraft and parts manufacturing, aircraft operation, aircraft airworthiness, and cabin safety. ASIs must complete a training program at the FAA Academy in Oklahoma City, Oklahoma, which includes airman evaluation and pilot testing techniques and procedures. ASIs also receive extensive on-the-job training and recurrent training on a regular basis. The FAA has approximately 3,700 inspectors located in its FSDO offices. All questions concerning pilot certification (and/or requests for other aviation information or services) should be directed to the local FSDO.

### **FAA Safety Team (FAASTeam)**

The FAA is dedicated to improving the safety of United States civilian aviation by conveying safety principles and practices through training, outreach, and education. The FAA



**Figure 1-13.** Atlanta Flight Standards District Office (FSDO).

Safety Team (FAASTeam) exemplifies this commitment. The FAASTeam has replaced the Aviation Safety Program (ASP), whose education of airmen on all types of safety subjects successfully reduced accidents. Its success led to its demise because the easy-to-fix accident causes have been addressed. To take aviation safety one step further, Flight Standards Service created the FAASTeam, which is devoted to reducing aircraft accidents by using a coordinated effort to focus resources on elusive accident causes.

Each of the FAA's nine regions has a Regional FAASTeam Office dedicated to this new safety program and managed by the Regional FAASTeam Manager (RFM). The FAASTeam is "teaming" up with individuals and the aviation industry to create a unified effort against accidents and tip the safety culture in the right direction. To learn more about this effort to improve aviation safety, to take a course at their online learning center, or to join the FAASTeam, visit their website at [www.faasafety.gov](http://www.faasafety.gov).

### **Obtaining Assistance from the FAA**

Information can be obtained from the FAA by phone, Internet/e-mail, or mail. To talk to the FAA toll-free 24 hours a day, call 1-866-TELL-FAA (1-866-835-5322). To visit the FAA's website, go to [www.faa.gov](http://www.faa.gov). Individuals can also e-mail an FAA representative at a local FSDO office by accessing the staff e-mail address available via the "Contact FAA" link at the bottom of the FAA home page. Letters can be sent to:

Federal Aviation Administration  
800 Independence Ave, SW  
Washington, DC 20591

### **FAA Reference Material**

The FAA provides a variety of important reference material for the student, as well as the advanced civil aviation pilot. In addition to the regulations provided online by the FAA, several other publications are available to the user. Almost all reference material is available online at [www.faa.gov](http://www.faa.gov) in downloadable format. Commercial aviation publishers also provide published and online reference material to further aid the aviation pilot.

### **Aeronautical Information Manual (AIM)**

The Aeronautical Information Manual (AIM) is the official guide to basic flight information and ATC procedures for the aviation community flying in the NAS of the United States. [Figure 1-14] An international version, containing parallel information as well as specific information on international airports, is also available. The AIM also contains information of interest to pilots, such as health and medical facts, flight

## Chapter 9

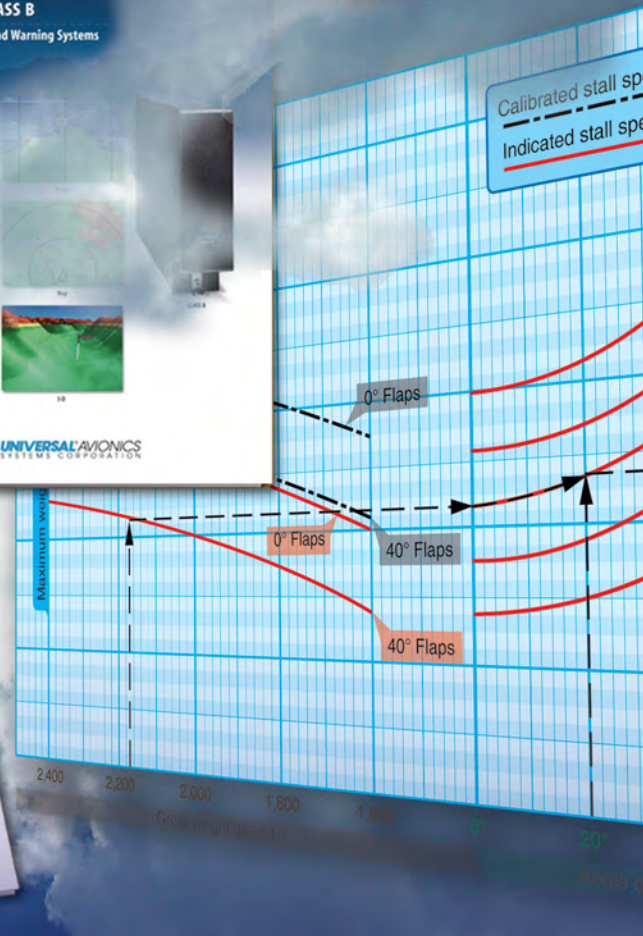
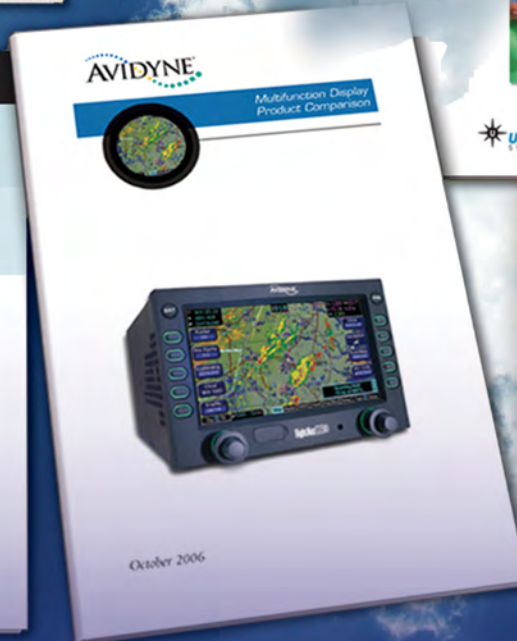
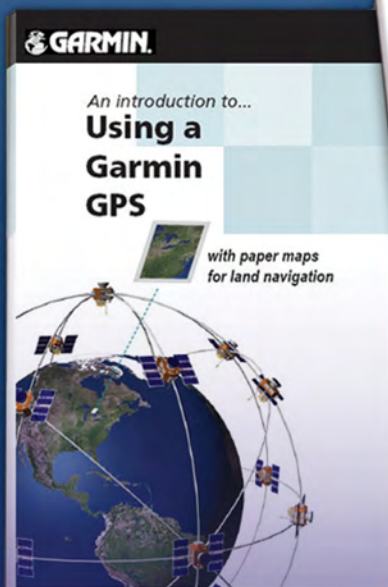
# Flight Manuals and Other Documents

### Introduction

Each aircraft comes with documentation and a set of manuals with which a pilot must be familiar in order to fly that aircraft. This chapter covers airplane flight manuals (AFM), the pilot's operating handbook (POH), and aircraft documents pertaining to ownership, airworthiness, maintenance, and operations with inoperative equipment. Knowledge of these required documents and manuals is essential for a pilot to conduct a safe flight.

### Airplane Flight Manuals (AFM)

Flight manuals and operating handbooks are concise reference books that provide specific information about a particular aircraft or subject. They contain basic facts, information, and/or instructions for the pilot about the operation of an aircraft, flying techniques, etc., and are intended to be kept on hand for ready reference.



The aircraft owner/information manual is a document developed by the aircraft manufacturer and contains general information about the make and model of the aircraft. The manual is not approved by the Federal Aviation Administration (FAA) and is not specific to an individual aircraft. The manual provides general information about the operation of an aircraft, is not kept current, and cannot be substituted for the AFM/POH.

An AFM is a document developed by the aircraft manufacturer and approved by the FAA. This book contains the information and instructions required to operate an aircraft safely. A pilot must comply with this information which is specific to a particular make and model of aircraft, usually by serial number. An AFM contains the operating procedures and limitations of that aircraft. Title 14 of the Code of Federal Regulations (14 CFR) part 91 requires that pilots comply with the operating limitations specified in the approved flight manuals, markings, and placards.

Originally, flight manuals followed whatever format and content the manufacturer felt was appropriate, but this changed with the acceptance of Specification No. 1 prepared by the General Aviation Manufacturers Association (GAMA). Specification No. 1 established a standardized format for all general aviation airplane and helicopter flight manuals.

The POH is a document developed by the aircraft manufacturer and contains FAA-approved AFM information. If “POH” is used in the main title, a statement must be included on the title page indicating that sections of the document are FAA approved as the AFM.

The POH for most light aircraft built after 1975 is also designated as the FAA-approved flight manual. The typical AFM/POH contains the following nine sections: General; Limitations; Emergency Procedures; Normal Procedures; Performance; Weight and Balance/Equipment List; Systems Description; Handling, Service, and Maintenance; and Supplements. Manufacturers also have the option of including additional sections, such as one on Safety and Operational Tips or an alphabetical index at the end of the POH.

### **Preliminary Pages**

While the AFM/POH may appear similar for the same make and model of aircraft, each manual is unique and contains specific information about a particular aircraft, such as the equipment installed and weight and balance information. Manufacturers are required to include the serial number and registration on the title page to identify the aircraft to which the manual belongs. If a manual does not indicate a specific aircraft registration and serial number, it is limited to general study purposes only.

Most manufacturers include a table of contents that identifies the order of the entire manual by section number and title. Usually, each section also contains a table of contents for that section. Page numbers reflect the section and page within that section (1-1, 1-2, 2-1, 3-1, etc.). If the manual is published in loose-leaf form, each section is usually marked with a divider tab indicating the section number, title, or both. The Emergency Procedures section may have a red tab for quick identification and reference.

### **General (Section 1)**

The General section provides the basic descriptive information on the airframe and powerplant(s). Some manuals include a three-dimensional drawing of the aircraft that provides dimensions of various components. Included are such items as wingspan, maximum height, overall length, wheelbase length, main landing gear track width, diameter of the rotor system, maximum propeller diameter, propeller ground clearance, minimum turning radius, and wing area. This section serves as a quick reference and helps a pilot become familiar with the aircraft.

The last segment of the General section contains definitions, abbreviations, explanations of symbology, and some of the terminology used in the POH. At the discretion of the manufacturer, metric and other conversion tables may also be included.

### **Limitations (Section 2)**

The Limitations section contains only those limitations required by regulation or that are necessary for the safe operation of the aircraft, powerplant, systems, and equipment. It includes operating limitations, instrument markings, color-coding, and basic placards. Some of the limitation areas are airspeed, powerplant, weight and loading distribution, and flight.

#### **Airspeed**

Airspeed limitations are shown on the airspeed indicator (ASI) by color coding and on placards or graphs in the aircraft. *[Figure 9-1]* A red line on the ASI shows the airspeed limit beyond which structural damage could occur. This is called the never-exceed speed ( $V_{NE}$ ). A yellow arc indicates the speed range between maximum structural cruising speed ( $V_{NO}$ ) and  $V_{NE}$ . Operation of an aircraft in the yellow airspeed arc is for smooth air only and then only with caution. A green arc depicts the normal operating speed range, with the upper end at  $V_{NO}$  and the lower end at stalling speed at maximum weight with the landing gear and flaps retracted ( $V_{S1}$ ). For airplanes, the flap operating range is depicted by the white arc, with the upper end at the maximum flap extended speed ( $V_{FE}$ ), and the lower end at the stalling speed with the landing gear and flaps in the landing configuration ( $V_{S0}$ ).





**Figure 9-1.** Single-engine airspeed indicator.

In addition to the markings listed above, small multi-engine airplanes have a red radial line to indicate single-engine minimum controllable airspeed ( $V_{MC}$ ). A blue radial line is used to indicate single-engine best rate of climb speed at maximum weight at sea level ( $V_{YSE}$ ). [Figure 9-2]

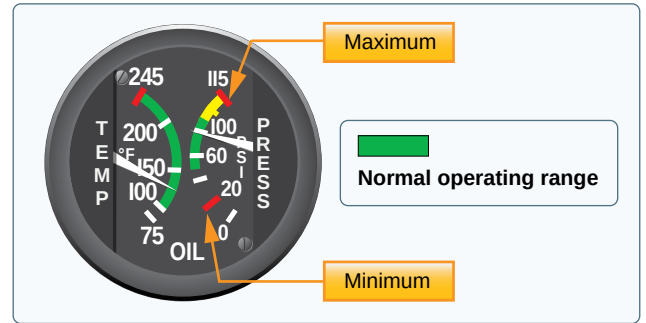
### Powerplant

The Powerplant Limitations portion describes operating limitations on an aircraft's reciprocating or turbine engine(s). These include limitations for takeoff power, maximum continuous power, and maximum normal operating power, which is the maximum power the engine can produce without any restrictions and is depicted by a green arc. Other items that can be included in this area are the minimum and maximum oil and fuel pressures, oil and fuel grades, and propeller operating limits. [Figure 9-3]

All reciprocating-engine powered aircraft must have a revolutions per minute (rpm) indicator for each engine. Aircraft equipped with a constant-speed propeller or rotor system use a manifold pressure gauge to monitor power output and a tachometer to monitor propeller or rotor speed. Both instruments depict the maximum operating limit with



**Figure 9-2.** Multi-engine airspeed indicator.

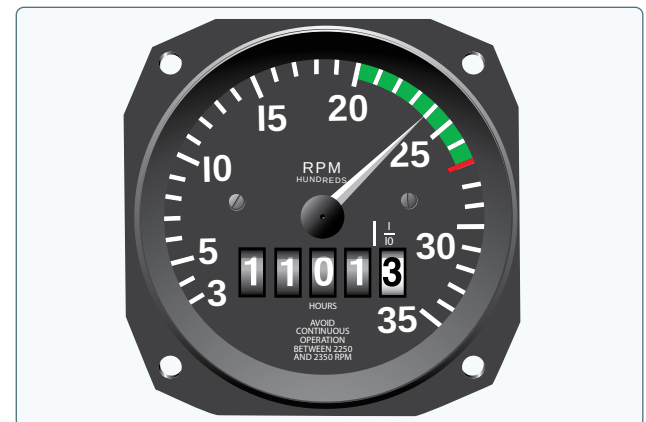


**Figure 9-3.** Minimum, maximum, and normal operating range markings on oil gauge.

a red radial line and the normal operating range with a green arc. [Figure 9-4] Some instruments may have a yellow arc to indicate a caution area.

### Weight and Loading Distribution

Weight and Loading Distribution contains the maximum certificated weights, as well as the center of gravity (CG) range. The location of the reference datum used in balance computations is included in this section. Weight and balance computations are not provided in this area, but rather in the weight and balance section of the AFM/POH.



**Figure 9-4.** Manifold pressure gauge (top) and tachometer (bottom).

## Flight Limits

Flight Limits list authorized maneuvers with appropriate entry speeds, flight load factor limits, and types of operation limits. It also indicates those maneuvers that are prohibited, such as spins or acrobatic flight, as well as operational limitations such as flight into known icing conditions.

## Placards

Most aircraft display one or more placards that contain information having a direct bearing on the safe operation of the aircraft. These placards are located in conspicuous places and are reproduced in the Limitations section or as directed by an Airworthiness Directive (AD). [Figure 9-5] Airworthiness Directives are explained in detail later in this chapter.

## Emergency Procedures (Section 3)

Checklists describing the recommended procedures and airspeeds for coping with various types of emergencies or critical situations are located in the Emergency Procedures section. Some of the emergencies covered include: engine failure, fire, and system failure. The procedures for inflight engine restarting and ditching may also be included. Manufacturers may first show an emergency checklist in an abbreviated form with the order of items reflecting the sequence of action. Amplified checklists that provide additional information on the procedures follow the abbreviated checklist. To be prepared for emergency situations, memorize the immediate action items and, after completion, refer to the appropriate checklist.



Figure 9-5. Placards are used to depict aircraft limitations.

Manufacturers may include an optional subsection entitled Abnormal Procedures. This subsection describes recommended procedures for handling malfunctions that are not considered emergencies.

## Normal Procedures (Section 4)

This section begins with a list of the airspeeds for normal operations. The next area consists of several checklists that may include preflight inspection, before starting procedures, starting engine, before taxiing, taxiing, before takeoff, climb, cruise, descent, before landing, balked landing, after landing, and post flight procedures. An Amplified Procedures area follows the checklists to provide more detailed information about the various previously mentioned procedures.

To avoid missing important steps, always use the appropriate checklists when available. Consistent adherence to approved checklists is a sign of a disciplined and competent pilot.

## Performance (Section 5)

The Performance section contains all the information required by the aircraft certification regulations and any additional performance information the manufacturer deems important to pilot ability to safely operate the aircraft. Performance charts, tables, and graphs vary in style, but all contain the same basic information. Examples of the performance information found in most flight manuals include a graph or table for converting calibrated airspeed to true airspeed; stall speeds in various configurations; and data for determining takeoff and climb performance, cruise performance, and landing performance. Figure 9-6 is an example of a typical performance graph. For more information on use of the charts, graphs, and tables, refer to Chapter 10, Aircraft Performance.

## Weight and Balance/Equipment List (Section 6)

The Weight and Balance/Equipment List section contains all the information required by the FAA to calculate the weight and balance of an aircraft. Manufacturers include sample weight and balance problems. Weight and balance is discussed in greater detail in Chapter 10, Weight and Balance.

## Systems Description (Section 7)

This section describes the aircraft systems in a manner appropriate to the pilot most likely to operate the aircraft. For example, a manufacturer might assume an experienced pilot will be reading the information for an advanced aircraft. For more information on aircraft systems, refer to Chapter 7, Aircraft Systems.

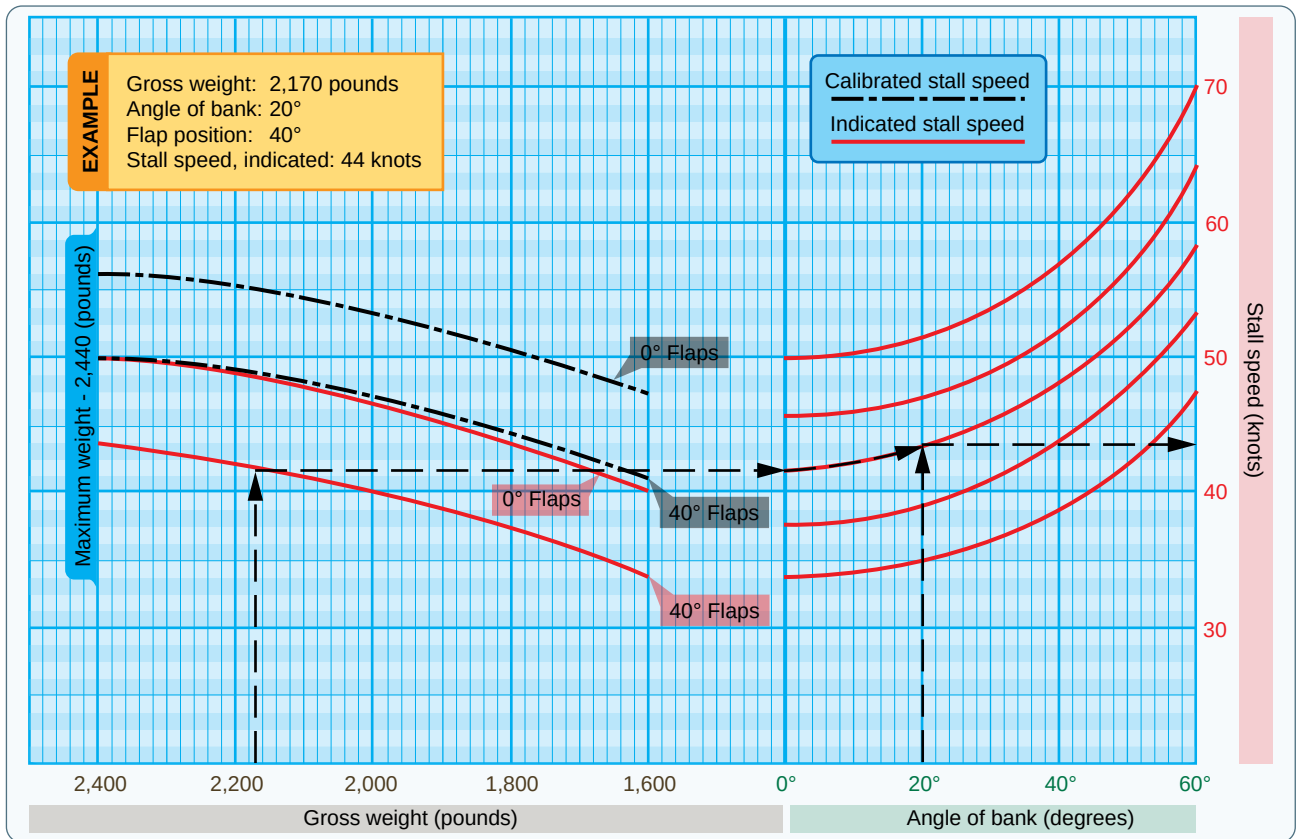


Figure 9-6. Stall speed chart.

### Handling, Service, and Maintenance (Section 8)

The Handling, Service, and Maintenance section describes the maintenance and inspections recommended by the manufacturer (and the regulations). Additional maintenance or inspections may be required by the issuance of ADs applicable to the airframe, engine, propeller, or components. This section also describes preventive maintenance that may be accomplished by certificated pilots, as well as the manufacturer's recommended ground handling procedures. It includes considerations for hangaring, tie-down, and general storage procedures for the aircraft.

### Supplements (Section 9)

The Supplements section contains information necessary to safely and efficiently operate the aircraft when equipped with optional systems and equipment (not provided with the standard aircraft). Some of this information may be supplied by the aircraft manufacturer or by the manufacturer of the optional equipment. The appropriate information is inserted into the flight manual at the time the equipment is installed. Autopilots, navigation systems, and air-conditioning systems are examples of equipment described in this section.

[Figure 9-7]



Figure 9-7. Supplements provide information on optional equipment.



## Safety Tips (Section 10)

The Safety Tips section is an optional section containing a review of information that enhances the safe operation of the aircraft. For example, physiological factors, general weather information, fuel conservation procedures, high altitude operations, or cold weather operations might be discussed.

## Aircraft Documents

### Certificate of Aircraft Registration

Before an aircraft can be flown legally, it must be registered with the FAA Aircraft Registry. The Certificate of Aircraft Registration, which is issued to the owner as evidence of the registration, must be carried in the aircraft at all times. [Figure 9-8]

The Certificate of Aircraft Registration cannot be used for operations when:

- The aircraft is registered under the laws of a foreign country

- The aircraft's registration is canceled upon written request of the certificate holder
- The aircraft is totally destroyed or scrapped
- The ownership of the aircraft is transferred
- The certificate holder loses United States citizenship

For additional information, see 14 CFR part 47, section 47.41. When one of the events listed in 14 CFR part 47, section 47.41 occurs, the previous owner must notify the FAA by filling in the back of the Certificate of Aircraft Registration, and mailing it to:

FAA Aircraft Registration Branch, AFS-750  
P.O. Box 25504  
Oklahoma City, OK 73125-0504

A dealer's aircraft registration certificate is another form of registration certificate, but is valid only for required flight tests by the manufacturer or in flights that are necessary for

| REGISTRATION NOT TRANSFERABLE                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| UNITED STATES OF AMERICA<br>DEPARTMENT OF TRANSPORTATION - FEDERAL AVIATION ADMINISTRATION<br>CERTIFICATE OF AIRCRAFT REGISTRATION                                                                                                                                                                                                                                                                                                            |                                                                                                                                   |
| NATIONALITY AND REGISTRATION MARKS <b>N48SB</b>                                                                                                                                                                                                                                                                                                                                                                                               | AIRCRAFT SERIAL NO. <b>9411</b>                                                                                                   |
| MANUFACTURER AND MANUFACTURER'S DESIGNATION OF AIRCRAFT<br><b>PITTS SIS</b>                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                   |
| ICAO Aircraft Address Code: <b>5163722</b>                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                   |
| ISSUED TO<br><b>JOHN SMITH<br/>111 WASHINGTON ROAD<br/>PORTLAND, OR 97111</b>                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                   |
| This certificate is issued for registration purposes only and is not a certificate of title. The Federal Aviation Administration does not determine rights of ownership as between private persons.                                                                                                                                                                                                                                           |                                                                                                                                   |
| It is certified that the above described aircraft has been entered on the register of the Federal Aviation Administration, United States of America, in accordance with the Convention on International Civil Aviation, dated December 7, 1944, and with the Federal Aviation Act of 1958, and regulations issued thereunder.                                                                                                                 |                                                                                                                                   |
| DATE OF ISSUE<br><b>JUNE 3, 1995</b>                                                                                                                                                                                                                                                                                                                                                                                                          | <i>David Benson</i> ADMINISTRATOR                                                                                                 |
| AC Form 8050-3(11/93) Supersedes previous editions                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                   |
| EFFECT OF REGISTRATION<br>Section 501(b) of the Federal Aviation Act of 1958 (49 U.S.C. 1401) provides: "...Registration shall not be evidence of ownership of aircraft in any proceeding in which such ownership by a particular person is, or may be, in issue." THIS CERTIFICATE MUST BE SIGNED AND RETURNED BY THE REGISTERED OWNER WITHIN 90 DAYS WHEN IT IS NO LONGER IN EFFECT FOR ANY REASON UNDER 14 C.F.R. 47.41(a)(1) THROUGH (9). |                                                                                                                                   |
| a. <input type="checkbox"/> Registration is cancelled at the request of the owner. (Also check and/or complete block b, c, d, e, or f).                                                                                                                                                                                                                                                                                                       | e. <input type="checkbox"/> The aircraft is to be registered under the laws of a foreign country: (NAME OF FOREIGN COUNTRY) _____ |
| b. <input type="checkbox"/> The aircraft is totally destroyed or scrapped.                                                                                                                                                                                                                                                                                                                                                                    | f. <input type="checkbox"/> The ownership of the aircraft is transferred to: (NAME) _____                                         |
| c. <input type="checkbox"/> United States citizenship has been lost, or the owner's status as a resident alien has changed (unless changed to that of a U.S. citizen).                                                                                                                                                                                                                                                                        | (ADDRESS) _____                                                                                                                   |
| d. <input type="checkbox"/> Thirty days have elapsed since the death of the registered owner (estate representative should sign).                                                                                                                                                                                                                                                                                                             | (CITY, STATE, ZIP) _____                                                                                                          |
| (SIGNATURE) _____                                                                                                                                                                                                                                                                                                                                                                                                                             | (TITLE) _____ (DATE) _____                                                                                                        |
| This certificate must be returned to:<br>AIRCRAFT REGISTRATION BRANCH, P.O. BOX 25504, OKLAHOMA CITY, OKLAHOMA 73125-0504                                                                                                                                                                                                                                                                                                                     |                                                                                                                                   |

U.S. Department of Transportation  
Federal Aviation Administration  
Office of Aviation System Standards  
P.O. Box 25504  
Oklahoma City, OK 73125-0504  
Official Business  
Penalty for Private Use \$300  
AC Form 8050-3(11/93) Supersedes previous editions  
TO: **JOHN SMITH  
111 WASHINGTON ROAD  
PORTLAND, OR 97111**

RETAIN THIS INFORMATION FOR FUTURE REFERENCE

CHANGE OF ADDRESS  
Federal Aviation Regulations require that the registered owner of the aircraft shall report in writing within 30 days any change in permanent mailing address. A revised Certificate of Registration will be issued without charge. The Application for Registration AC Form 8050-1 (1-92) be used to report a change of address.

REPLACEMENT OF CERTIFICATE  
If this certificate is lost, destroyed, or mutilated, a replacement may be obtained at the written request of the holder. Send your request and \$2.00 (check or money order made payable to United States Treasury) to:  
Aircraft Registration Branch  
P.O. Box 25504  
Oklahoma City, Oklahoma 73125-0504

NOTE: All correspondence should include the registration "N" number, manufacturer, model, and serial number of the aircraft.

Figure 9-8. AC Form 8050-3, Certificate of Aircraft Registration.

the sale of the aircraft by the manufacturer or a dealer. The dealer must remove the certificate when the aircraft is sold.

Upon complying with 14 CFR part 47, section 47.31, the pink copy of the application for an Aircraft Registration Application, Aeronautical Center (AC) Form 8050-1, provides authorization to operate an unregistered aircraft for a period not to exceed 90 days. Since the aircraft is unregistered, it cannot be operated outside of the United States until a permanent Certificate of Aircraft Registration is received and placed in the aircraft.

The FAA does not issue any certificate of ownership or endorse any information with respect to ownership on a Certificate of Aircraft Registration.

NOTE: For additional information concerning the Aircraft Registration Application or the Aircraft Bill of Sale, contact the nearest FAA Flight Standards District Office (FSDO).

### Airworthiness Certificate

An Airworthiness Certificate is issued by a representative of the FAA after the aircraft has been inspected, is found to meet the requirements of 14 CFR part 21, and is in condition for safe operation. The Airworthiness Certificate must be displayed in the aircraft so it is legible to the passengers and crew whenever it is operated. The Airworthiness Certificate must remain with the aircraft unless it is sold to a foreign purchaser.

A Standard Airworthiness Certificate is issued for aircraft type certificated in the normal, utility, acrobatic, commuter, transport categories, and manned free balloons. *Figure 9-9* illustrates a Standard Airworthiness Certificate, and an explanation of each item in the certificate follows.

1. **Nationality and Registration Marks.** The “N” indicates the aircraft is registered in the United States. Registration marks consist of a series of up to five numbers or numbers and letters. In this case, N2631A is the registration number assigned to this aircraft.
2. **Manufacturer and Model.** Indicates the manufacturer, make, and model of the aircraft.
3. **Aircraft Serial Number.** Indicates the manufacturer’s serial number assigned to the aircraft, as noted on the aircraft data plate.
4. **Category.** Indicates the category in which the aircraft must be operated. In this case, it must be operated in accordance with the limitations specified for the “NORMAL” category.
5. **Authority and Basis for Issuance.** Indicates the aircraft conforms to its type certificate and is considered in condition for safe operation at the time of inspection and issuance of the certificate. Any exemptions from the applicable airworthiness standards are briefly noted here and the exemption number given. The word “NONE” is entered if no exemption exists.

| UNITED STATES OF AMERICA<br>DEPARTMENT OF TRANSPORTATION-FEDERAL AVIATION ADMINISTRATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                    |                                             |                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------|------------------------------------|
| <b>STANDARD AIRWORTHINESS CERTIFICATE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                    |                                             |                                    |
| 1 NATIONALITY AND<br>REGISTRATION MARKS<br><b>N12345</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 MANUFACTURER AND MODEL<br><b>Douglas DC-6A</b>   | 3 AIRCRAFT SERIAL<br>NUMBER<br><b>43219</b> | 4 CATEGORY<br><b>Transport</b>     |
| 5 AUTHORITY AND BASIS FOR ISSUANCE<br>This airworthiness certificate is issued pursuant to the Federal Aviation Act of 1958 and certifies that, as of the date of issuance, the aircraft to which issued has been inspected and found to conform to the type certificate therefor, to be in condition for safe operation, and has been shown to meet the requirements of the applicable comprehensive and detailed airworthiness code as provided by Annex 8 to the Convention on International Civil Aviation, except as noted herein.<br>Exceptions:<br><br><b>None</b> |                                                    |                                             |                                    |
| 6 TERMS AND CONDITIONS<br>Unless sooner surrendered, suspended, revoked, or a termination date is otherwise established by the Administrator, this airworthiness certificate is effective as long as the maintenance, preventative maintenance, and alterations are performed in accordance with Parts 21, 43, and 91 of the Federal Aviation Regulations, as appropriate, and the aircraft is registered in the United States.                                                                                                                                           |                                                    |                                             |                                    |
| DATE OF ISSUANCE<br><b>01/20/00</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FAA REPRESENTATIVE<br><b>E.R. White E.R. White</b> |                                             | DESIGNATION NUMBER<br><b>NE-XX</b> |
| Any iteration, reproduction, or misuse of this certificate may be punishable by a fine not exceeding \$1,000 or imprisonment not exceeding 3 years or both.<br>THIS CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT IN ACCORDANCE WITH APPLICABLE FEDERAL AVIATION REGULATIONS.                                                                                                                                                                                                                                                                                             |                                                    |                                             |                                    |
| FAA Form 8100-2 (04-11) Supersedes Previous Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                    |                                             |                                    |

**Figure 9-9.** FAA Form 8100-2, Standard Airworthiness Certificate.

6. *Terms and Conditions.* Indicates the Airworthiness Certificate is in effect indefinitely if the aircraft is maintained in accordance with 14 CFR parts 21, 43, and 91, and the aircraft is registered in the United States.

Also included are the date the certificate was issued and the signature and office identification of the FAA representative.

A Standard Airworthiness Certificate remains in effect if the aircraft receives the required maintenance and is properly registered in the United States. Flight safety relies in part on the condition of the aircraft, which is determined by inspections performed by mechanics, approved repair stations, or manufacturers that meet specific requirements of 14 CFR part 43.

A Special Airworthiness Certificate is issued for all aircraft certificated in other than the Standard classifications, such as Experimental, Restricted, Limited, Provisional, and Light-Sport Aircraft (LSA). LSA receive a pink special airworthiness certificate; however, there are exceptions. For example, the Piper Cub is in the LSA category, but it was certificated as a normal aircraft during its manufacture. When purchasing an aircraft classified as other than Standard, it is recommended that the local FSDO be contacted for an explanation of the pertinent airworthiness requirements and the limitations of such a certificate.

### **Aircraft Maintenance**

Maintenance is defined as the preservation, inspection, overhaul, and repair of an aircraft, including the replacement of parts. Regular and proper maintenance ensures that an aircraft meets an acceptable standard of airworthiness throughout its operational life.

Although maintenance requirements vary for different types of aircraft, experience shows that aircraft need some type of preventive maintenance every 25 hours of flying time or less and minor maintenance at least every 100 hours. This is influenced by the kind of operation, climatic conditions, storage facilities, age, and construction of the aircraft. Manufacturers supply maintenance manuals, parts catalogs, and other service information that should be used in maintaining the aircraft.

### **Aircraft Inspections**

Under 14 CFR part 91, the primary responsibility for maintaining an aircraft in an airworthy condition falls on the owner or operator of the aircraft. Certain inspections must be performed on the aircraft, and the owner must maintain the airworthiness of the aircraft during the time between required inspections by having any defects corrected.

Under 14 CFR, part 91, subpart E, all civil aircraft are required to be inspected at specific intervals to determine the overall condition. The interval depends upon the type of operations in which the aircraft is engaged. All aircraft need to be inspected at least once every 12 calendar months, while inspection is required for others after every 100 hours of operation. Some aircraft are inspected in accordance with an inspection system set up to provide for total inspection of the aircraft on the basis of calendar time, time in service, number of system operations, or any combination of these.

All inspections should follow the current manufacturer's maintenance manual, including the Instructions for Continued Airworthiness concerning inspection intervals, parts replacement, and life-limited items as applicable to the aircraft.

### **Annual Inspection**

Any reciprocating engine or single-engine turbojet/turbopropeller-powered small aircraft (weighing 12,500 pounds or less) flown for business or pleasure and not flown for compensation or hire is required to be inspected at least annually. The inspection shall be performed by a certificated airframe and powerplant (A&P) mechanic who holds an inspection authorization (IA) by the manufacturer of the aircraft or by a certificated and appropriately rated repair station. The aircraft may not be operated unless the annual inspection has been performed within the preceding 12 calendar months. A period of 12 calendar months extends from any day of a month to the last day of the same month the following year. An aircraft overdue for an annual inspection may be operated under a Special Flight Permit issued by the FAA for the purpose of flying the aircraft to a location where the annual inspection can be performed. However, all applicable ADs that are due must be complied with before the flight.

### **100-Hour Inspection**

All aircraft under 12,500 pounds (except turbojet/turbopropeller-powered multi-engine airplanes and turbine powered rotorcraft), used to carry passengers for hire, must receive a 100-hour inspection within the preceding 100 hours of time in service and must be approved for return to service. Additionally, an aircraft used for flight instruction for hire, when provided by the person giving the flight instruction, must also have received a 100-hour inspection. This inspection must be performed by an FAA-certificated A&P mechanic, an appropriately rated FAA-certificated repair station, or by the aircraft manufacturer. An annual inspection, or an inspection for the issuance of an Airworthiness Certificate, may be substituted for a required 100-hour inspection. The 100-hour limitation may be exceeded by no more than 10



hours for the purpose of traveling to a location at which the required inspection can be performed. Any excess time used for this purpose must be included in computing the next 100 hours of time in service.

### **Other Inspection Programs**

The annual and 100-hour inspection requirements do not apply to large (over 12,500 pounds) airplanes, turbojets, or turbopropeller-powered multi-engine airplanes or to aircraft for which the owner complies with a progressive inspection program. Details of these requirements may be determined by referencing 14 CFR, part 43, section 43.11 and 14 CFR part 91, subpart E, or by inquiring at a local FSDO.

### ***Altimeter System Inspection***

Under 14 CFR, part 91, section 91.411, requires that the altimeter, encoding altimeter, and related system must be tested and inspected within the 24 months prior to operating in controlled airspace under instrument flight rules (IFR). This applies to all aircraft being operated in controlled airspace.

### ***Transponder Inspection***

Title 14 CFR, part 91, section 91.413, requires that before a transponder can be used under 14 CFR, part 91, section 91.215(a), it shall be tested and inspected within the 24 months prior to operation of the aircraft regardless of airspace restrictions.

### ***Emergency Locator Transmitter***

An emergency locator transmitter (ELT) is required by 14 CFR, part 91, section 91.207, and must be inspected within 12 calendar months after the last inspection for the following:

- Proper installation
- Battery corrosion
- Operation of the controls and crash sensor
- The presence of a sufficient signal radiated from its antenna

The ELT must be attached to the airplane in such a manner that the probability of damage to the transmitter in the event of crash impact is minimized. Fixed and deployable automatic type transmitters must be attached to the airplane as far aft as practicable. Batteries used in the ELTs must be replaced (or recharged, if the batteries are rechargeable):

- When the transmitter has been in use for more than 1 cumulative hour
- When 50 percent of the battery useful life or, for rechargeable batteries, 50 percent of useful life of the charge has expired

An expiration date for replacing (or recharging) the battery must be legibly marked on the outside of the transmitter and entered in the aircraft maintenance record. This does not apply to batteries that are essentially unaffected during storage intervals, such as water-activated batteries.

### ***Preflight Inspections***

The preflight inspection is a thorough and systematic means by which a pilot determines if an aircraft is airworthy and in condition for safe operation. POHs and owner/information manuals contain a section devoted to a systematic method of performing a preflight inspection.

### **Minimum Equipment Lists (MEL) and Operations With Inoperative Equipment**

Under 14 CFR, all aircraft instruments and installed equipment are required to be operative prior to each departure. When the FAA adopted the minimum equipment list (MEL) concept for 14 CFR part 91 operations, it allowed aircraft to be operated with inoperative equipment determined to be nonessential for safe flight. At the same time, it allowed part 91 operators, without an MEL, to defer repairs on nonessential equipment within the guidelines of part 91.

The FAA has two acceptable methods of deferring maintenance on small rotorcraft, non-turbine powered airplanes, gliders, or lighter-than-air aircraft operated under part 91. They are the deferral provision of 14 CFR, part 91, section 91.213(d) and an FAA-approved MEL.

The deferral provision of 14 CFR, part 91, section 91.213(d) is widely used by most pilot/operators. Its popularity is due to simplicity and minimal paperwork. When inoperative equipment is found during a preflight inspection or prior to departure, the decision should be to cancel the flight, obtain maintenance prior to flight, or to defer the item or equipment.

Maintenance deferrals are not used for inflight discrepancies. The manufacturer's AFM/POH procedures are to be used in those situations. The discussion that follows assumes that the pilot wishes to defer maintenance that would ordinarily be required prior to flight.

Using the deferral provision of 14 CFR, part 91, section 91.213(d), the pilot determines whether the inoperative equipment is required by type design, 14 CFR, or ADs. If the inoperative item is not required, and the aircraft can be safely operated without it, the deferral may be made. The inoperative item shall be deactivated or removed and an INOPERATIVE placard placed near the appropriate switch, control, or indicator. If deactivation or removal involves

maintenance (removal always will), it must be accomplished by certificated maintenance personnel and recorded in accordance with 14 CFR part 43.

For example, if the position lights (installed equipment) were discovered to be inoperative prior to a daytime flight, the pilot would follow the requirements of 14 CFR, part 91, section 91.213(d).

The deactivation may be a process as simple as the pilot positioning a circuit breaker to the OFF position or as complex as rendering instruments or equipment totally inoperable. Complex maintenance tasks require a certificated and appropriately rated maintenance person to perform the deactivation. In all cases, the item or equipment must be placarded INOPERATIVE.

All small rotorcraft, non-turbine powered airplanes, gliders, or lighter-than-air aircraft operated under 14 CFR part 91 are eligible to use the maintenance deferral provisions of 14 CFR, part 91, section 91.213(d). However, once an operator requests an MEL, and a Letter of Authorization (LOA) is issued by the FAA, then the use of the MEL becomes mandatory for that aircraft. All maintenance deferrals must be accomplished in accordance with the terms and conditions of the MEL and the operator-generated procedures document.

The use of an MEL for an aircraft operated under 14 CFR part 91 also allows for the deferral of inoperative items or equipment. The primary guidance becomes the FAA-approved MEL issued to that specific operator and N-numbered aircraft.

The FAA has developed master minimum equipment lists (MMELs) for aircraft in current use. Upon written request by an operator, the local FSDO may issue the appropriate make and model MMEL, along with an LOA, and the preamble. The operator then develops operations and maintenance (O&M) procedures from the MMEL. This MMEL with O&M procedures now becomes the operator's MEL. The MEL, LOA, preamble, and procedures document developed by the operator must be on board the aircraft during each operation. The FAA considers an approved MEL to be a supplemental type certificate (STC) issued to an aircraft by serial number and registration number. It, therefore, becomes the authority to operate that aircraft in a condition other than originally type certificated.

With an approved MEL, if the position lights were discovered inoperative prior to a daytime flight, the pilot would make an entry in the maintenance record or discrepancy record provided for that purpose. The item would then either be repaired or deferred in accordance with the MEL. Upon

confirming that daytime flight with inoperative position lights is acceptable in accordance with the provisions of the MEL, the pilot would leave the position lights switch OFF, open the circuit breaker (or whatever action is called for in the procedures document), and placard the position light switch as INOPERATIVE.

There are exceptions to the use of the MEL for deferral. For example, should a component fail that is not listed in the MEL as deferrable (the tachometer, flaps, or stall warning device, for example), then repairs are required to be performed prior to departure. If maintenance or parts are not readily available at that location, a special flight permit can be obtained from the nearest FSDO. This permit allows the aircraft to be flown to another location for maintenance. This allows an aircraft that may not currently meet applicable airworthiness requirements, but is capable of safe flight, to be operated under the restrictive special terms and conditions attached to the special flight permit.

Deferral of maintenance is not to be taken lightly, and due consideration should be given to the effect an inoperative component may have on the operation of an aircraft, particularly if other items are inoperative. Further information regarding MELs and operations with inoperative equipment can be found in AC 91-67, Minimum Equipment Requirements for General Aviation Operations Under FAR Part 91.

## Preventive Maintenance

Preventive maintenance is regarded as simple or minor preservation operations and the replacement of small standard parts, not involving complex assembly operations. Allowed items of preventative maintenance are listed and limited to the items of 14 CFR part 43, appendix A(c).

### *Maintenance Entries*

All pilots who perform preventive maintenance must make an entry in the maintenance record of the aircraft. The entry must include the following information:

1. A description of the work, such as "changed oil (Shell Aero-50) at 2,345 hours"
2. The date of completion of the work performed
3. The pilot's name, signature, certificate number, and type of certificate held

### *Examples of Preventive Maintenance*

The following examples of preventive maintenance are taken from 14 CFR, part 43, Maintenance, Preventive Maintenance, Rebuilding, and Alteration, which should be consulted for a more in-depth look at the preventive maintenance a pilot can perform on an aircraft. Remember, preventive maintenance

is limited to work that does not involve complex assembly operations including the following:

- Removal, installation, and repair of landing gear tires and shock cords; servicing landing gear shock struts by adding oil, air, or both; servicing gear wheel bearings; replacing defective safety wiring or cotter keys; lubrication not requiring disassembly other than removal of nonstructural items, such as cover plates, cowlings, and fairings; making simple fabric patches not requiring rib stitching or the removal of structural parts or control surfaces. In the case of balloons, the making of small fabric repairs to envelopes (as defined in, and in accordance with, the balloon manufacturer's instructions) not requiring load tape repair or replacement.
- Replenishing hydraulic fluid in the hydraulic reservoir; refinishing decorative coating of fuselage, balloon baskets, wings, tail group surfaces (excluding balanced control surfaces), fairings, cowlings, landing gear, cabin, or flight deck interior when removal or disassembly of any primary structure or operating system is not required; applying preservative or protective material to components where no disassembly of any primary structure or operating system is involved and where such coating is not prohibited or is not contrary to good practices; repairing upholstery and decorative furnishings of the cabin, flight deck, or balloon basket interior when the repair does not require disassembly of any primary structure or operating system or interfere with an operating system or affect the primary structure of the aircraft; making small, simple repairs to fairings, nonstructural cover plates, cowlings, and small patches and reinforcements not changing the contour to interfere with proper air flow; replacing side windows where that work does not interfere with the structure or any operating system, such as controls, electrical equipment, etc.
- Replacing safety belts, seats or seat parts with replacement parts approved for the aircraft, not involving disassembly of any primary structure or operating system, bulbs, reflectors, and lenses of position and landing lights.
- Replacing wheels and skis where no weight-and-balance computation is involved; replacing any cowling not requiring removal of the propeller or disconnection of flight controls; replacing or cleaning spark plugs and setting of spark plug gap clearance; replacing any hose connection, except hydraulic connections; however, prefabricated fuel lines may be replaced.
- Cleaning or replacing fuel and oil strainers or filter elements; servicing batteries, cleaning balloon burner pilot and main nozzles in accordance with the balloon manufacturer's instructions.
- The interchange of balloon baskets and burners on envelopes when the basket or burner is designated as interchangeable in the balloon type certificate data and the baskets and burners are specifically designed for quick removal and installation; adjustment of nonstructural standard fasteners incidental to operations.
- The installations of anti-misfueling devices to reduce the diameter of fuel tank filler openings only if the specific device has been made a part of the aircraft type certificate data by the aircraft manufacturer, the aircraft manufacturer has provided FAA-approved instructions for installation of the specific device, and installation does not involve the disassembly of the existing tank filler opening; troubleshooting and repairing broken circuits in landing light wiring circuits.
- Removing and replacing self-contained, front instrument panel-mounted navigation and communication devices that employ tray-mounted connectors which connect the unit when the unit is installed into the instrument panel (excluding automatic flight control systems, transponders, and microwave frequency distance measuring equipment (DME)). The approved unit must be designed to be readily and repeatedly removed and replaced, and pertinent instructions must be provided. Prior to the unit's intended use, an operational check must be performed in accordance with the applicable sections of 14 CFR part 91 on checking, removing, and replacing magnetic chip detectors.
- Inspection and maintenance tasks prescribed and specifically identified as preventive maintenance in a primary category aircraft type certificate or STC holder's approved special inspection and preventive maintenance program when accomplished on a primary category aircraft.
- Updating self-contained, front instrument panel-mounted air traffic control (ATC) navigational software databases (excluding those of automatic flight control systems, transponders, and microwave frequency DME), only if no disassembly of the unit is required and pertinent instructions are provided; prior to the unit's intended use, an operational check must be performed in accordance with applicable sections of 14 CFR part 91.



Certificated pilots, excluding student pilots, sport pilots, and recreational pilots, may perform preventive maintenance on any aircraft that is owned or operated by them provided that the aircraft is not used in air carrier service and does not qualify under 14 CFR parts 121, 129, or 135. A pilot holding a sport pilot certificate may perform preventive maintenance on an aircraft owned or operated by that pilot if that aircraft is issued a special airworthiness certificate in the LSA category. (Sport pilots operating LSA should refer to 14 CFR part 65 for maintenance privileges.) 14 CFR part 43, appendix A, contains a list of the operations that are considered to be preventive maintenance.

### Repairs and Alterations

Repairs and alterations are classified as either major or minor. 14 CFR part 43, appendix A, describes the alterations and repairs considered major. Major repairs or alterations shall be approved for return to service on FAA Form 337, Major Repair and Alteration, by an appropriately rated certificated repair station, an FAA-certificated A&P mechanic holding an IA, or a representative of the Administrator. Minor repairs and minor alterations may be approved for return to service with a proper entry in the maintenance records by an FAA-certificated A&P mechanic or an appropriately certificated repair station.

For modifications of experimental aircraft, refer to the operating limitations issued to that aircraft. Modifications in accordance with FAA Order 8130.2, Airworthiness Certification of Aircraft and Related Products, may require the notification of the issuing authority.

### Special Flight Permits

A special flight permit is a Special Airworthiness Certificate authorizing operation of an aircraft that does not currently meet applicable airworthiness requirements but is safe for a specific flight. Before the permit is issued, an FAA inspector may personally inspect the aircraft or require it to be inspected by an FAA-certificated A&P mechanic or an appropriately certificated repair station to determine its safety for the intended flight. The inspection shall be recorded in the aircraft records.

The special flight permit is issued to allow the aircraft to be flown to a base where repairs, alterations, or maintenance can be performed; for delivering or exporting the aircraft; or for evacuating an aircraft from an area of impending danger. A special flight permit may be issued to allow the operation of an overweight aircraft for flight beyond its normal range over water or land areas where adequate landing facilities or fuel is not available.

If a special flight permit is needed, assistance and the necessary forms may be obtained from the local FSDO

or Designated Airworthiness Representative (DAR).  
[Figure 9-10]

## Airworthiness Directives (ADs)

A primary safety function of the FAA is to require correction of unsafe conditions found in an aircraft, aircraft engine, propeller, or appliance when such conditions exist and are likely to exist or develop in other products of the same design. The unsafe condition may exist because of a design defect, maintenance, or other causes. Airworthiness Directives (ADs), under 14 CFR, part 39, define the authority and responsibility of the Administrator for requiring the necessary corrective action. ADs are used to notify aircraft owners and other interested persons of unsafe conditions and to specify the conditions under which the product may continue to be operated. ADs are divided into two categories:

1. Those of an emergency nature requiring immediate compliance prior to further flight
2. Those of a less urgent nature requiring compliance within a specified period of time

ADs are regulatory and shall be complied with unless a specific exemption is granted. It is the responsibility of the aircraft owner or operator to ensure compliance with all pertinent ADs, including those ADs that require recurrent or continuing action. For example, an AD may require a repetitive inspection each 50 hours of operation, meaning the particular inspection shall be accomplished and recorded every 50 hours of time in service. Owners/operators are reminded that there is no provision to overfly the maximum hour requirement of an AD unless it is specifically written into the AD. To help determine if an AD applies to an amateur-built aircraft, contact the local FSDO.

14 CFR, part 91, section 91.417 requires a record to be maintained that shows the current status of applicable ADs, including the method of compliance; the AD number and revision date, if recurring; next due date and time; the signature; type of certificate; and certificate number of the repair station or mechanic who performed the work. For ready reference, many aircraft owners have a chronological listing of the pertinent ADs in the back of their aircraft, engine, and propeller maintenance records.

All ADs and the AD Biweekly are free on the Internet at <http://rgl.faa.gov> and are available through e-mail. Individuals can enroll for the e-mail service at the website above. Paper copies of the Summary of Airworthiness Directives and the AD Biweekly may be purchased from the Superintendent of Documents. The Summary contains all the valid ADs previously published and is divided into

|                                                                                                                                        |                                                                                                                                                                                                                                                                                            |                                            |                                    |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|
| UNITED STATES OF AMERICA<br>DEPARTMENT OF TRANSPORTATION - FEDERAL AVIATION ADMINISTRATION<br><b>SPECIAL AIRWORTHINESS CERTIFICATE</b> |                                                                                                                                                                                                                                                                                            |                                            |                                    |
| <b>A</b>                                                                                                                               | CATEGORY/DESIGNATION      Special Flight Permit                                                                                                                                                                                                                                            |                                            |                                    |
|                                                                                                                                        | PURPOSE      Production Flight Testing or Customer Demonstration                                                                                                                                                                                                                           |                                            |                                    |
| <b>B</b>                                                                                                                               | MANU-FACTURER                                                                                                                                                                                                                                                                              | NAME      The Boeing Company               |                                    |
|                                                                                                                                        |                                                                                                                                                                                                                                                                                            | ADDRESS      P.O. Box 767, Renton WA 13567 |                                    |
| <b>C</b>                                                                                                                               | FLIGHT                                                                                                                                                                                                                                                                                     | FROM      N/A                              |                                    |
|                                                                                                                                        |                                                                                                                                                                                                                                                                                            | TO      N/A                                |                                    |
| <b>D</b>                                                                                                                               | N- N/A                                                                                                                                                                                                                                                                                     |                                            | SERIAL NO.      N/A                |
|                                                                                                                                        | BUILDER      N/A                                                                                                                                                                                                                                                                           |                                            | MODEL      N/A                     |
| <b>E</b>                                                                                                                               | DATE OF ISSUANCE      01/31/2001                                                                                                                                                                                                                                                           |                                            | EXPIRY      01/31/2001             |
|                                                                                                                                        | OPERATING LIMITATIONS DATED 01/31/2001 ARE PART OF THIS CERTIFICATE                                                                                                                                                                                                                        |                                            |                                    |
|                                                                                                                                        | SIGNATURE OF FAA REPRESENTATIVE<br>Sam T. Smith<br><i>Sam T. Smith</i>                                                                                                                                                                                                                     |                                            | DESIGNATION OR OFFICE NO.<br>NM-XX |
|                                                                                                                                        | Any alteration, reproduction or misuse of this certificate may be punishable by a fine not exceeding \$1,000 or imprisonment not exceeding 3 years, or both. THIS CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT IN ACCORDANCE WITH APPLICABLE TITLE 14, CODE OF FEDERAL REGULATIONS (CFR). |                                            |                                    |

FAA Form 8130-7 (07/04) SEE REVERSE SIDE

**Figure 9-10.** FAA Form 8130-7, Special Airworthiness Certificate.

two areas. The small aircraft and helicopter books contain all ADs applicable to small aircraft (12,500 pounds or less maximum certificated takeoff weight) and ADs applicable to all helicopters. The large aircraft books contain all ADs applicable to large aircraft.

For current information on how to order paper copies of AD books and the AD Biweekly, visit the FAA online regulatory and guidance library at: <http://rgl.faa.gov>.

### Aircraft Owner/Operator Responsibilities

The registered owner/operator of an aircraft is responsible for:

- Having a current Airworthiness Certificate and a Certificate of Aircraft Registration in the aircraft.
- Maintaining the aircraft in an airworthy condition, including compliance with all applicable ADs and assuring that maintenance is properly recorded.
- Keeping abreast of current regulations concerning the operation and maintenance of the aircraft.
- Notifying the FAA Aircraft Registry immediately of any change of permanent mailing address, of the sale or export of the aircraft, or of the loss of the eligibility to register an aircraft. (Refer to 14 CFR, part 47, section 47.41.)

- Having a current Federal Communications Commission (FCC) radio station license if equipped with radios, including emergency locator transmitter (ELT), if operated outside of the United States.

### Chapter Summary

Knowledge of an aircraft's AFM/POH and documents, such as ADs, provide pilots with ready access to pertinent information needed to safely fly a particular aircraft. By understanding the operations, limitations, and performance characteristics of the aircraft, the pilot can make educated flight decisions. By learning what preventive maintenance is allowed on the aircraft, a pilot can maintain his or her aircraft in an airworthy condition. The goal of every pilot is a safe flight. Flight manuals and aircraft documentation are essential tools used to reach that goal.

